

# Notes on Alfaro et al. (2024)

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**Motivation** 1. Empirics: Causal effect of second-order shock is correlated with the first-order in economic down-turn; 2. Theory: Current models have difficulty explaining the slow recovery after the Great Recession.

**Model Overview** DSGE with heterogeneous firms. 1) Real friction: Investment has fixed cost; 2) Financial friction: Liquidity management by saving cash; 3) Stochasticity: Uncertainty and financing costs.

Results: 1. Amplification effect with financial friction: Double the negative impact of uncertainty shocks on investment; 2. Persistence effect with financial friction: Double the duration of the drop; 3. Propagation effect onto financial variables: Affect both investment and cash (accumulation) and equity payout (drop).

**Empirical Results** **TO DO!**

## Model

- (a) Technology: Firms use physical capital to produce a single final good with aggregate productivity  $X_t$  and firm-specific productivity  $z_{j,t}$ :

$$y_{j,t} = X_t z_{j,t} k_{j,t}^\alpha.$$

Productivity terms follow  $AR(1)$  process:

$$\begin{aligned}\log X_{t+1} &= (1 - \rho^X) \log \bar{X} + \rho^X \log X_t + \sigma_t^X \epsilon_{t+1}^X \\ \log z_{j,t+1} &= \rho^z \log z_{j,t} + \sigma_t^z \epsilon_{j,t+1}^z,\end{aligned}$$

where  $\epsilon_{t+1}^X$  and  $\epsilon_{j,t+1}^z$  are i.i.d. standard normal shocks and  $\sigma_t^X$  and  $\sigma_t^z$  are macro and micro uncertainty (time-varying conditional volatilities) of the productivity processes.<sup>1</sup> These uncertainties follow a two-state Markov process with high and low states.

Capital accumulation is standard with non-convex adjustment costs.

- (b) Cash accumulation: Cash investment and return to holding cash (question mark here).
- (c) External financing: Exist only when payout is negative. External financing condition is a two-state Markov process with high and low states as a multiplier to the required financing.
- (d) Firm's problem: Maximize payout minus external financing cost with continuation value.
- (e) Households: Have access to equity.
- (f) Equilibrium: Goods market, equity market, and money market (money supply is exogenous, equal the demand from firms).

## References

**Alfaro, Iván, Nicholas Bloom, and Xiaoji Lin**, “The Finance Uncertainty Multiplier,” *Journal of Political Economy*, February 2024, 132 (2), 577–615.

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<sup>1</sup>In my paper, the modelling of uncertainty is exactly à la this.